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Experimental study of methane reforming with products of complete methane combustion in a reformer filled with a nickel-based catalyst

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A R T I C L E I N F O

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A B S T R A C T

An experimental investigation of methane reforming with flue gases obtained by combustion of methane with an excess air factor of 1 is carried out over a nickel-based catalyst. A set of experiments was conducted to understand the effects of operational parameters such as temperature, pressure, residence time and inlet gas composition on the conversion of the reactants. In addition, two shapes of catalysts such as a solid cylinder and ring were used in experiments. It is found that at the residence time above 4 kg_{cat} s/mol the methane conversion has a value close to equilibrium. A change in pressure from 1 to 5 bar has a negligible effect on methane conversion, not more than 1–2% for various operating parameters. It has been experimentally determined that the use of an extended surface catalyst gives a higher methane conversion at a fixed catalyst weight. Therefore, for a fixed residence time, the methane conversion over a Raschig rings is 1–2% higher than over a solid cylinder. The steam addition to the inlet reaction gas mixture increases the methane conversion significantly; especially at the temperature range from 500 to 800 °C.

1. Introduction

Reforming of hydrocarbons with various oxidizing agents, such as steam AND carbon dioxide, is widespread in various industries. It is known that steam reforming of natural gas is the main process for the large-scale production of hydrogen [1] and synthesis gas [2]. Dry reforming of natural gas is widely used to produce synthesis gas with a high carbon monoxide-to-hydrogen (CO/H_2) ratio [3]. Also, combined steam and dry reforming of methane (the main component of natural gas (up to 98–99%)) is used in the chemical process industry to produce synthesis gas with a given composition [4].

In recent years, many researchers and scientists from different countries have paid considerable attention to the use of methane reforming to improve the energy efficiency of fuel-consuming plants: the industrial gas furnaces [5], the gas turbines [6], the internal combustion engines [7]. Such a method for increasing energy efficiency is called thermochemical recuperation. The main concept of thermochemical recuperation (TCR) is the transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low-heating value [8]. The possibility of using TCR for many fuel-consuming plants increases the importance of researching various aspects of hydrocarbon reforming.

In the literature, a large number of studies devoted to the study of methane reforming by various oxidants have been published. In general, scientific publications are devoted to the design and performance study of the new catalysts for both steam methane reforming (SMR) [9] and dry methane reforming (DMR) [10], to the energy and exergy analysis of SMR or DMR for the different schemes [11], to the numerical modeling of methane reforming processes [12].

Among the main disadvantages of steam and carbon dioxide reforming of methane can be identified the need for the production of oxidants and their heating to high temperatures. For example, steam production for steam reforming of methane requires more than 10% of the heat spent on the process [13]. One solution to this problem is the use of flue gases as methane oxidants, which contain steam and carbon dioxide. This method of methane reforming has attracted interest from both industrial and environmental perspectives. Firstly, from an environmental point of view, the two most abundant carbon-containing greenhouse gases, methane and carbon dioxide, can be utilized effectively in this reaction and converted into useful chemical products. Secondly, at present, this methane reforming method has found wide application for the thermochemical heat recovery system.

Combined steam-dry methane reforming was studied both theoretically and experimentally. Ozkara-Aydınoglu conducted a thermodynamic analysis of combined steam-dry reforming of methane to produce synthesis gas at the wide range of operational parameters such as temperature, pressure, and inlet methane-steam-carbon dioxide ratio ($CH_4/H_2O/CO_2$) [14]. The author has established that the maximum

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methane conversion is achieved in the temperature range of 900–1000 K for different initial compositions and pressures.

Al-Nakoua and El-Naas reported results of a numerical investigation of combined steam-dry methane reforming in a fluidized bed membrane reactor [15]. They described the influence of key parameters on reactor performance including reactor temperature, reactor pressure, steam to methane feed ratio (S/C) and carbon dioxide to methane feed ratio (CO_2/C).

Lemonidou and Vasalos experimentally investigated combined CO_2 methane reforming with steam methane reforming over 5 wt.% Ni/CaO-Al_2O_3 (a gamma aluminum oxide support) catalyst [16]. They investigated steam addition to the carbon dioxide methane reforming process to test the increase of the conversion of methane and the decrease of carbon deposition in a catalyst. Industrially, both steam and dry methane reforming are mostly performed over nickel supported Al_2O_3 catalyst around 1000–1100 K. However, the catalyst is mostly deactivated by carbon deposition onto its surface, which is the major problem to catalyst stability. Khania and co-authors presented article with results of experimental investigation of $ZnLaAlO_4$ supported nickel (Ni), platinum (Pt) and ruthenium (Ru) nanocatalysts applied for combined dry-steam reforming of methane [17].

At the same time, the available publications do not cover all the practically interesting cases dealing with the reforming of methane by the products of its complete combustion (flue gases). Most of the works devoted to the experimental study of reforming used pure steam and carbon dioxide in the presence of nitrogen as oxidants. For example, to conduct engineering calculations of the thermochemical waste-heat recuperation systems, it is necessary to have accurate data on methane reforming with flue gases in order to design and optimize thermochemical reactors, as well as perform an energy analysis of the TCR systems.

In the presented article, an experimental study of methane reforming with flue gases (obtained by the complete combustion of methane) is carried out for various operating parameters such as temperature, pressure, and $CH_4/CO_2/H_2O$ inlet composition, as well as the residence time. Kinetic data were obtained, which made it possible to establish the dependence of the degree of methane conversion on residence time for different temperatures, pressures, and inlet gas compositions. In this study, flue gases as the methane oxidant were obtained by methane combustion at an excess air ratio of 1.0.

2. Experiments

A schematic diagram of the experimental setup is presented in Fig. 1. The experimental setup can be divided into three blocks. The first block is the block for preparing of the reaction mixture. The second block is the block of the reforming process. The third block is the block for measuring and storing of the results. The main elements of the first block are an electrical steam generator and a gas burner. The main element of the second block is a methane reformer. The main component of the third block is a portable gas analyzer.

2.1. Experimental setup

The experimental setup mainly consists of the following elements: 1 – a methane reformer; 2 – an electrical muffle furnace; 3 – a watt-hour meter; 4 – the thermocouples; 5 – a portable gas analyzer; 6 – a personal computer; 7 – the mass-flow meters; 8 – a tangential burner; 9 – an electrical steam generator; 10 – a gas intercooler; 11 – a manometer.

The methane reformer is made of high-alloy steel and placed in an electric muffle furnace. The electrical furnace heats the reaction space up to 1000 °C. The thermocouples are made of a tungsten-rhenium alloy. The gas composition is controlled by the portable gas analyzer Gasboard 3100, which can be connected to a personal computer. In a leak-proof gas burner with a tangential admission of methane, methane combustion takes place with an excess air factor equal to unity. Superheated steam is produced in the electric steam generator PG-LEK 1.1 (in Russian). High purity methane is used as a fuel and as a reagent for reaction reforming in the experiments, as well as a high purity compressed air is used for combustion. High purity distilled water is used for a steam production.

The methane and compressed air were obtained from the cylinders. After pressure reduction by mean of a regulator, the flow rate of methane and air were controlled by the mass-flow meters. Steam and flue gas are mixed with methane in a predetermined ratio. This homogeneous mixture of methane-flue gas-steam was then passed into the reactor.

A photograph of the experimental setup and some elements of the schematic diagram are shown in Fig. 2. The numbers of the elements in Fig. 2 coincide with the figure caption to Fig. 1.

2.2. Catalyst

The reaction space of the methane reformer is filled with Ni-Al_2O_3 catalyst with the different catalyst particle shapes. For experiment the Ni-Al_2O_3 catalyst is used with the following chemical composition: nickel oxide – NiO = 14.5%; silicon oxide – SiO_2 = 0.2%; support CaO-MgO-La_2O_3-Al_2O_3 [18]. Other properties are presented below: bulk density is 680 kg/m^3; surface area is 4.5 m^2/g_{cat}; average porosity of catalyst particles is about 41%. Two types of the particle shapes are

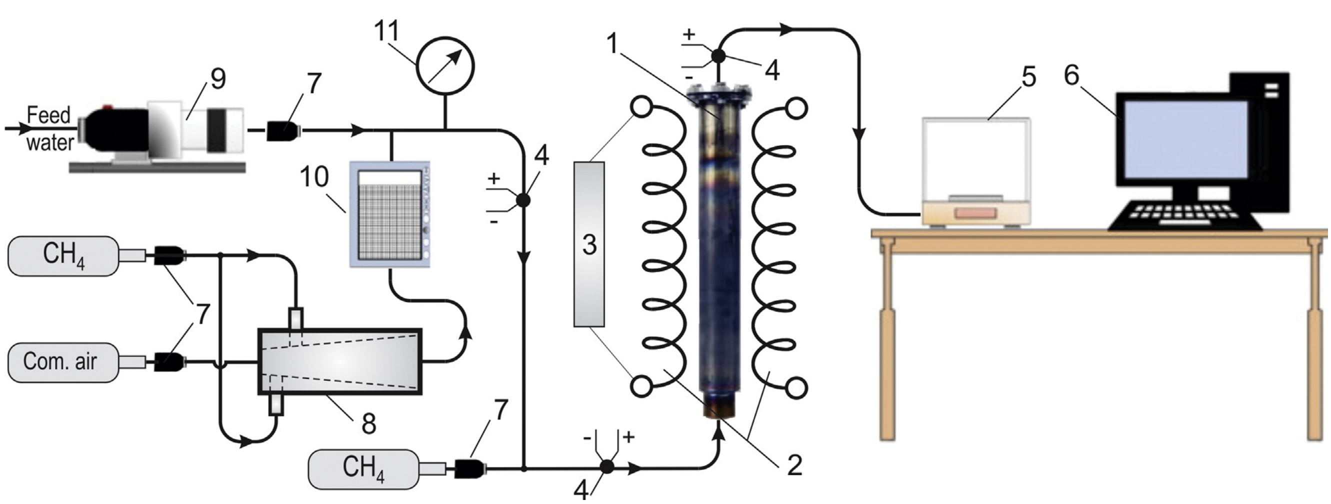

Fig. 1. Schematic view of experimental setup.

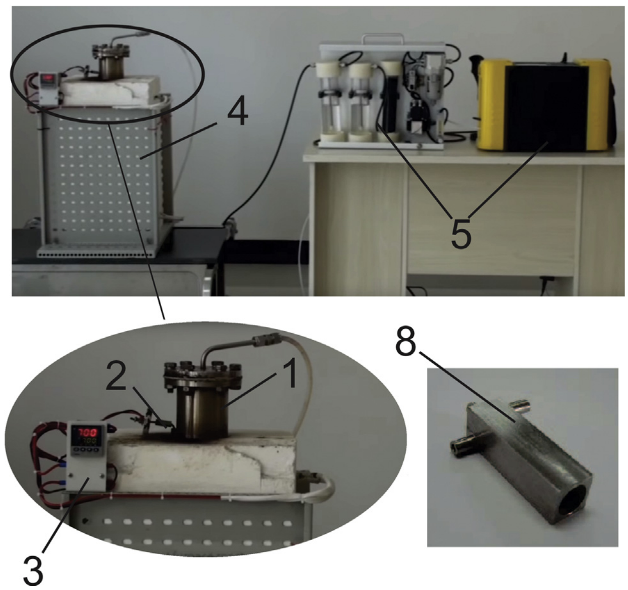

Fig. 2. Photograph of experimental setup.

used for the experiments (left side of Fig. 3): A – a simple cylinder; B – a Raschig ring (cylinder with one hole on axial line).

The catalyst particles are placed into the tube with a diameter of 100 mm. The catalysts layer depth is 600 mm. The geometrical characteristics of catalysts are shown in Table 1.

Scanning Electron Microscopy (SEM) image of the catalyst is presented in Fig. 3 (right side). SEM photo of catalyst structure is provided by the catalyst manufacturer NIAP-CATALYZATOR LLC (Novomoskovsk, Russia). The author thanks the manufacturer NIAP-CATALYZATOR LLC for the SEM image.

2.3. Experimental procedure

In the present study, a set of experiments was performed for various operational parameters: a temperature range from 400 to 800 °C, a pressure from 1 to 5 bar, a composition of the inlet reaction mixture, a residence time (residence time i.e. catalyst pellet weight divided by the feed rate of the methane expressed as kg_{cat} s/mol shows mass of catalyst bed per mole flow-rate on methane inlet). The residence time was varied due to a change in the flow rate of the reaction mixture. A set of experiments was performed when steam is added to the flue gases.

Before the experiment, the electrical furnace heats the catalyst to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. Heating of the catalyst takes approximately 15–17 min. During this period, the temperature in the reformer stabilizes throughout the reaction space. In parallel with the heating of the catalyst, combustion of methane is regulated by means of flowmeters. The temperature of the combustion products is regulated in the heat exchanger to a temperature that is higher than the catalyst temperature in the reformer. This is due to the fact that when the methane for reforming reaction is fed into the combustion products, the temperature of the mixture will decrease. The target temperature of the flue gases after the burner is determined experimentally in such a way that the temperature of the reaction mixture before the reformer is equal to the temperature of the reaction space of the reformer.

At the initial time, the system of flue gas supply to the reaction space is not connected to the reformer. After the reformer becomes warmed up and the combustion of methane is regulated, the flue gas supply system is connected to the reformer. With the help of an electric heater, the temperature in the reaction space during the experiments is regulated. Each experiment has a duration of 10–15 min. Fresh catalyst is used for each experiment. During the experiment, the composition of the reformed products and the heat consumption for the reforming reaction are fixed every minute.

3. Results and discussion

To analyze the effectiveness of the methane reforming process, the conversions of reactants and the yield of main products of reformate, both hydrogen and carbon monoxide, are defined as follows:

– methane conversion:

$$X_{CH_4} = \frac{CH_{4(in)} - CH_{4(out)}}{CH_{4(in)}}. \tag{1}$$

– carbon dioxide conversion:

$$X_{CO_2} = \frac{CO_{2(in)} - CO_{2(out)}}{CO_{2(in)}} \tag{2}$$

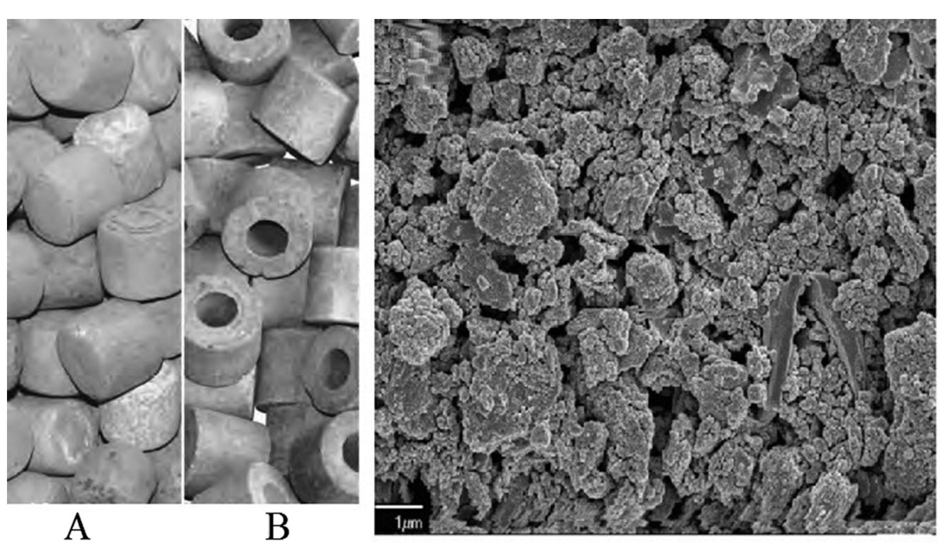

Fig. 3. Left side are photographs of the packed beds filled by the catalysts with different shapes: A – a cylindrical; B – a Raschig ring. Right side is SEM image for Ni-Al_2O_3 catalyst.

Table 1
The geometrical characteristics of catalyst.

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diam.	Geom.surface, m^2/m^3	Interparticle porosity m^3/m^3
A	25	25	–	182	0.35–0.38
B	25	25	1x13	347	0.46–0.49

– steam conversion:

$$X_{H_2O} = \frac{H_2O_{(in)} - H_2O(out)}{H_2O(in)} \tag{3}$$

– hydrogen yield:

$$Y_{H_2} = \frac{2 \cdot H_2(out)}{4 \cdot CH_{4(in)} + 2 \cdot H_{2,(in)}} \tag{4}$$

– carbon monoxide yield:

$$Y_{H_2} = \frac{CO_{(out)}}{CO_{2(in)} + CH_{4(in)}} \tag{5}$$

For further analysis, the parameter β is used. Here, β is the flue gases obtained by combustion of 1 mol of methane at excess air factor of 1.0. With complete combustion of 1 mol of methane, beta is equal to: $\beta = 2H_2O + CO_2 + 7.52N_2$.

3.1. Methane conversion

The maximum methane conversion can be achieved at equilibrium conditions. The equilibrium conversion of methane depends on the temperature, pressure, and composition of the inlet reaction mixture, and is independent of the type of catalyst, residence time and other process parameters. To estimate conversion of methane obtained in the experiments, the equilibrium conversion of methane is calculated by using Gibbs free energy minimization method. For the thermodynamic analysis, a commercial software ASPEN-HYSYS is used. The algorithm of calculation the equilibrium gas composition is presented in the author's previous work [19].

Fig. 4 shows changes of the methane conversions as a function of temperature when a various composition of inlet reaction mixture is fed into the methane reformer at the pressure of 1 bar for the different residence times. Raschig rings are used as a catalyst. Solid lines show the approximate curves of the experimental results. Hollow figures show the experimental results. Two compositions of the reaction mixture were investigated CH_4: $\frac{1}{3}\beta = CH_4{:}0.67H_2O{:}0.33CO_2{:}2.51\,N_2$ and CH_4: $\frac{2}{3}\beta = CH_4{:}1.34H_2O{:}0.66CO_2{:}5.02\,N_2$. For each case, the residence time was chosen as close as possible to each other, however, due to of the increase of the mass flow rate of the reaction mixture through the reformer, minor discrepancies were observed.

Fig. 4 shows the positive effect of temperature on CH_4 conversion. As the temperature increases, the methane conversion increases for each residence time. When the residence time is low, the methane conversion is almost proportional to temperature. However, the effect of temperature on CH_4 conversion has a non-linear relationship at the high residence time. Moreover, the addition of extra flue gas to the feed stream increased the methane conversion considerably by 6–11% for the different temperatures. For a low residence time, the methane conversion tends to an equilibrium value. The equilibrium CH_4 conversion obtained in this study correlates well with the results of the thermodynamic analysis of the combined steam-dry methane reforming presented by Özkara-Aydınoğlu [14].

The methane conversion also depends on the shape of the catalyst, as shown in Fig. 5. Raschig rings give a higher methane conversion than a solid cylinder. The catalytic activity of the surface for these catalysts is equal. Therefore, the efficiency of the Raschig rings can be explained by the fact that in this case the geometric surface of the catalyst bed is increased.

Fig. 6 depicts the methane conversion as a function of the residence time at different temperature, CH_4: $\frac{1}{3}\beta$, catalyst shape – a Raschig ring, pressure 1 bar. Dashed lines denote the equilibrium methane conversion for each temperature. It is seen that at the fixed temperatures, the methane conversion increase with increasing residence time. However, the curves also show that as the residence time increases to higher than 4 kg_{cat} s/mol in the temperature range higher 700 °C, the methane conversion increases slightly. Obviously, with a further increase in the residence time, the methane conversion reaches a maximum and stable value because the methane reforming process reaches to close to an equilibrium condition.

For engineering calculations of the methane reforming process, it is convenient to use expressions in which the methane conversion depends on the residence time. A set of such expressions for various pressures, temperatures, compositions of the initial mixture, and the shape of the catalyst was obtained by regression analysis of the experimental data. Second and third-degree polynomial regression was chosen correlating relationships between the methane conversion and residence time for a fixed pressure, temperature, inlet gas composition, catalyst shape. Using polynomial regression to the experimental results for the methane reforming process, the relationships between the

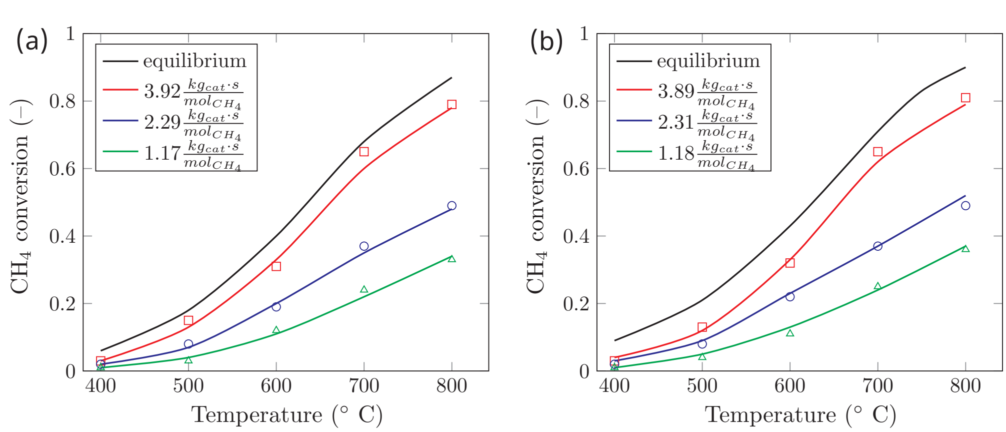

Fig. 4. Effect of temperature on the methane conversions at p = 1 bar for the various inlet gas compositions: (a) CH_4: $\frac{1}{3}\beta$; (b) CH_4: $\frac{2}{3}\beta$.

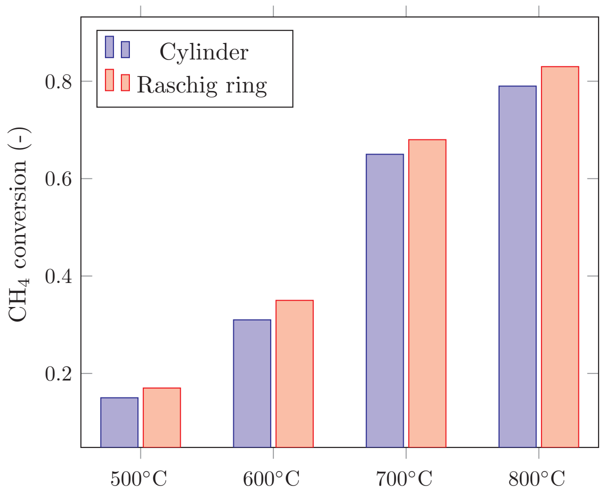

Fig. 5. Effect of catalyst shape on the methane conversions at p = 1 bar for the inlet gas composition as CH_4: $\frac{1}{3}\beta$.

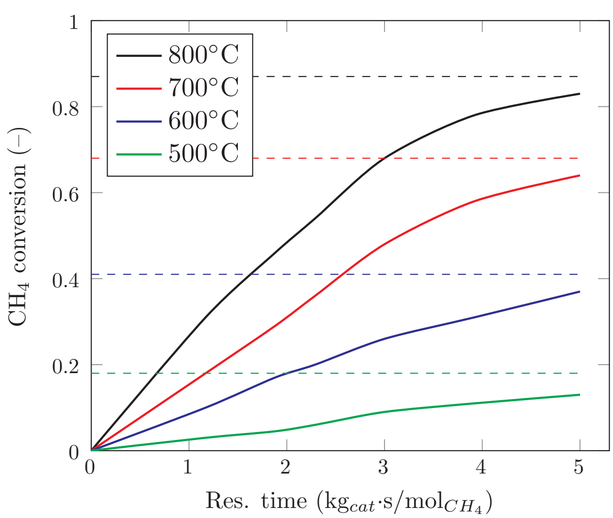

Fig. 6. Methane conversion as a function of residence time at different temperature, CH_4: $\frac{1}{3}\beta$, catalyst shape –Raschig ring, pressure 1 bar.

methane conversion and residence time for a fixed pressure, temperature, inlet gas composition, catalyst shape for a fixed temperature, pressure and the ratio of methane:flue gas were established as follows:

$$X_{CH_4} = a_0 + a_1\left(\frac{W}{F_{CH_4}}\right) + a_2\left(\frac{W}{F_{CH_4}}\right)^2 + a_3\left(\frac{W}{F_{CH_4}}\right)^3, \quad (6)$$

where a_i – correlation coefficients; W – mass of catalyst, kg; F_{CH_4} – molar flow rate of CH_4, mol/s.

The results of the regression analysis are presented in Table 2. In Table 2 the value of the correlation coefficients for CH_4: $\frac{2}{3}\beta$, where the cylinder catalyst is used, are approximately equal to the value where the Raschig ring catalyst is used for CH_4: $\frac{1}{3}\beta$. Therefore, these values are not shown in Table 2.

To assess the accuracy of the approximation expressions in Table 2, an analysis of the error distribution is performed. A histogram of the error distribution of the approximating function for CH_4: $\frac{1}{3}\beta$, p = 1 bar, cylinder, T = 700 °C within the residence time from 0 to 4 kg_{cat} s/mol is shown in Fig. 7.

The statistical analysis by the Pearson criterion, performed on the basis of the 9-bit histogram in Fig. 7, showed that the actual error distribution of the approximating function for CH_4: $\frac{1}{3}\beta$, p = 1 bar, cylinder, T = 700 °C within the residence time from 0 to 4 kg_{cat} s/mol is in

Table 2
Correlation coefficients for Eq. (6).

	T, °C	a_0	a_1	a_2	a_3
CH_4: $\frac{1}{3}\beta$ p = 1 bar cylinder	800	−0.0061	0.3128	−0.0327	0.0008
	700	−0.0078	0.1737	−0.0019	−0.0013
	600	0.1906	−0.0809	0.0442	−0.0041
CH_4: $\frac{1}{3}\beta$ p = 1 bar Raschig ring	800	−0.0023	−0.0105	0.2757	0.0033
	700	−0.0083	0.1890	−0.0054	−0.0011
	600	0.1962	−0.0822	0.0466	−0.0045
CH_4: $\frac{2}{3}\beta$ p = 1 bar Raschig ring	800	−0.0033	0.3602	−0.0455	0.0018
	700	−0.0218	0.1354	−0.0242	0.0305
	600	0.2170	−0.0735	−0.0735	−0.0041

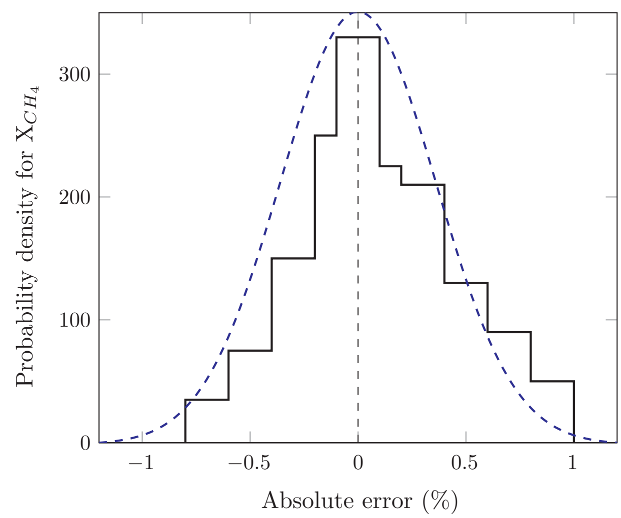

Fig. 7. Histogram of error distribution of the approximating function for CH_4: $\frac{1}{3}\beta$, p = 1 bar, cylinder, T = 700 °C within the residence time from 0 to 4 kg_{cat} s/mol: dashed line – Gaussian curve; solid line – histogram.

good agreement with the law of the normal Gaussian distribution (χ_{func} = 4.09 < χ_{exp} = 12.592, with a confidence level of P = 95% and the number of degrees of freedom m = 6). Therefore, for engineering calculations with high accuracy, it is possible to use approximation dependences, given in Table 2.

Table 3 shows the changes in the steam and carbon dioxide conversions for the methane reforming process at a fixed CH_4: $\frac{1}{3}\beta$ ratio. It is seen that keeping the methane-to-oxidant ratio at 1.0 the reactant conversion tends to unity with increasing temperature. In addition, the use of Raschig rings gives a higher conversion due to more efficient use of the catalyst mass. However, in practice, the use of methane-to-oxidant ratio equal to 1.0 is difficult, because even with an experiment duration of about 15 min, traces of soot carbon are observed on the surface of the catalyst after experiments. The carbon formation in methane reforming by flue gases is the subject of a separate author's study.

3.2. Hydrogen and carbon monoxide yields

The main components of the products of the methane reforming process are hydrogen and carbon monoxide. The H_2 and CO yields for the methane reforming process are presented in Fig. 8. For the all inlet gas compositions, increasing temperature led to an increase in the hydrogen yield. However, upon addition of the flue gas to the feedstock, the trends changed. If the hydrogen yield monotone increase with an increase of temperature for the all residence time range, then for the high temperature and the high residence time the hydrogen yield

Table 3
CO_2 and H_2O conversion at 3.92 kg_{cat} s/mol for p = 1 bar and CH_4: $\frac{1}{3}\beta$.

Catalyst shape	CO_2 conversion				H_2O conversion			
	500 °C	600 °C	700 °C	800 °C	500 °C	600 °C	700 °C	800 °C
Equilibrium	1.8	10.2	56.3	88.6	32.4	56.0	71.1	89.5
Cylinder	1.1	8.6	51.2	84.3	27.1	52.2	67.2	84.9
Raschig ring	1.3	8.9	51.8	85.1	27.9	52.7	68.0	86.1

reaches a maximum value and then decreases with increasing temperature. A maximum of H_2 yield for CH_4: $\frac{2}{3}\beta$ is reached around temperature of 700–800 °C for the high residence time. A decrease in hydrogen yield at high temperature may be two reasons. Firstly, at high temperatures dry methane reforming reaction, which yields less hydrogen compared to steam methane reforming, and water-gas shift reaction, which simultaneously consumes hydrogen, are favoured. Secondly, the addition flue gases over the stoichiometric amount of methane, steam and carbon dioxide in the inlet reaction mixture are shown in the denominator of the hydrogen yield equation Eq. (3), which also lowered the resulting hydrogen yield values. In conclusion, all these results indicate that proper control of the residence time and the inlet gas mixture and temperature is essential to obtain the maximum hydrogen yield.

Fig. 9 shows the dependence of the carbon monoxide yield on temperature. With increasing temperature, the CO yield increases monotonically for all cases. However, with the addition of additional flue gas to the inlet reaction mixture, the CO yield is reduced. Firstly, this is due to the fact that CO_2 is added in the denominator in Eq. (3) for a fixed amount of methane, while the amount of CO_2 is above the stoichiometric one. Secondly, there is an additional steam in the inlet reaction mixture, which consumes CO for the water-gas shift reaction.

3.3. Effect of pressure

According to the Le Chatelier-Brown principle, the methane reforming process with steam and carbon dioxide is affected not only by the feed gas composition and the temperature but also by pressure. This is due to the fact that both steam and carbon dioxide reforming of methane proceeds with an increase in the volume of the reaction products. It is can be seen from chemical equations of the steam methane reforming and dry methane reforming. For 2 mol of the inlet reaction mixture, the products of reactions are containing 4 mol.

$$CH_4 + H_2O \Leftrightarrow CO + H_2 + 206.1\ \text{kJ/mol}; \tag{7}$$

$$CH_4 + CO_2 \Leftrightarrow 2CO + H_2 + 247.3\ \text{kJ/mol}. \tag{8}$$

Using a leak-proof burner with a tangential supply of methane allows obtaining the methane combustion at various pressures. Therefore, the effect of pressure on the methane reforming process has been studied.

In this study, the methane reforming process has been investigated for the high residence time 3.92 kg_{cat} s/mol at the pressure of 1, 3 and 5 bar for the stoichiometric methane-to-oxidant ratio CH_4: $\frac{1}{3}\beta$. The use of high-temperature combustion products as an oxidizing agent of methane is possible only with moderately low pressure. This is due to the fact that the compression of flue gases at high temperatures is a technically complex process. In addition, as indicated in the introduction, the results of this study are intended to be used for the development of thermochemical heat recovery systems [20]. In such systems, it is supposed to use flue gases with moderately low pressure (before 5 bar). Fig. 10 shows the effect of pressure on the methane conversions as a function of temperature with CH_4: $\frac{1}{3}\beta$ when the catalyst such as Raschig rings is used. Fig. 10 presents that the methane conversions are suppressed by increasing pressure within the temperature range under study. With increasing pressure, the methane conversion decreases because the total product mole number is larger than that of the inlet reactants. In addition, as shown by Xu and Froment, the rate of the conversion reaction is inversely proportional to the pressure [21]. However, the effect of pressure is not more than 1–1.8% and in engineering calculations, the effect of pressure at a moderately low pressure (below 5 bar) can be considered insignificant.

Moreover, the effect of pressure on CH_4 conversion can be explained by diffusion. The diffusion rate of the reaction products depends on the concentration and partial pressure. Diffusion depends on pressure and temperature according to the expression:

$$D_i = D_0\left(\frac{T}{T_0}\right)^m \cdot \frac{P_0}{P}; \tag{9}$$

where D_0 – diffusion at temperature of T_0 and pressure of P_0;D_i – diffusion at temperature of T and pressure of P; m – correlation coefficient.

According to Eq. (9), with increasing pressure, diffusion decreases. However, the effect of diffusion also decreases due to moderately high velocity of the reaction mixture. Therefore, a slight difference in the methane conversion at different pressures is determined primarily by the mechanism of chemical reactions of steam-dry methane reforming.

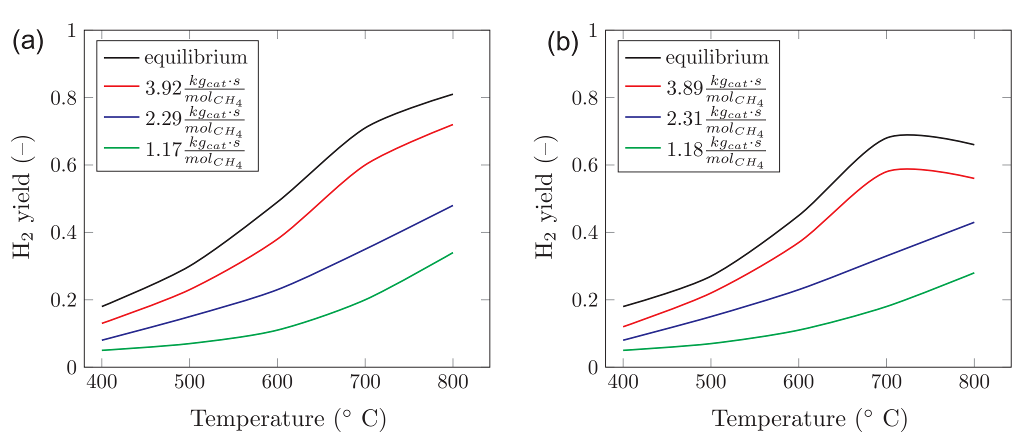

Fig. 8. Effect of temperature on the hydrogen yield at p = 1 bar for the various inlet gas compositions: (a) CH_4: $\frac{1}{3}\beta$; (b) CH_4: $\frac{2}{3}\beta$.

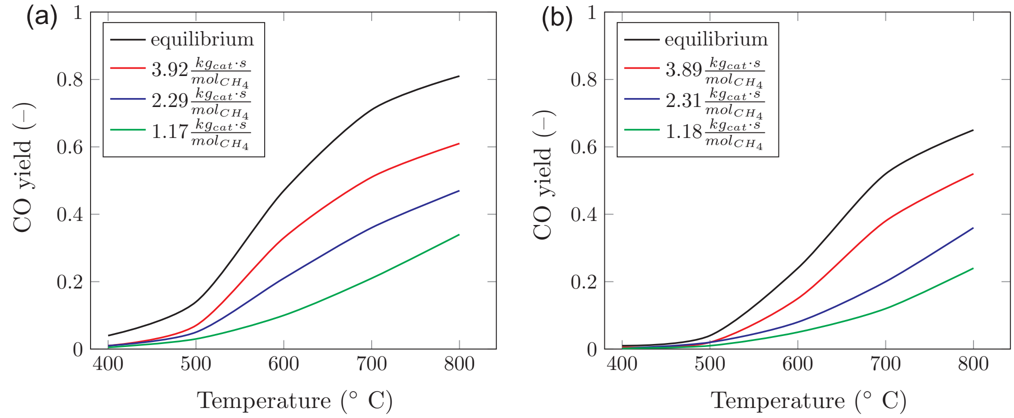

Fig. 9. Effect of temperature on the carbon monoxide yield at p = 1 bar for the various inlet gas compositions: (a) CH_4: $\frac{1}{3}\beta$; (b) CH_4: $\frac{2}{3}\beta$.

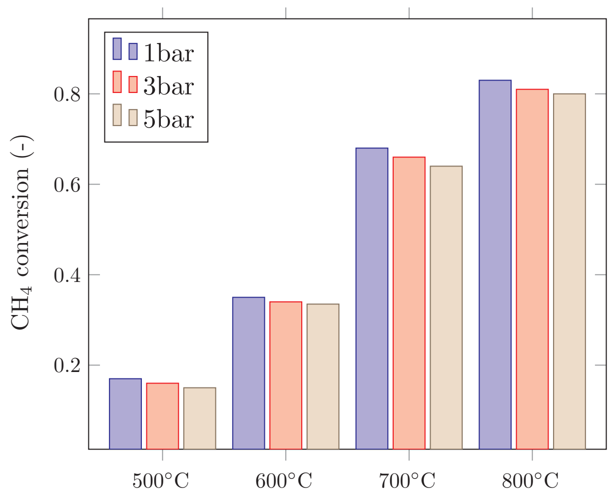

Fig. 10. Effect of pressure on the methane conversion at CH_4: $\frac{1}{3}\beta$ for the various temperatures.

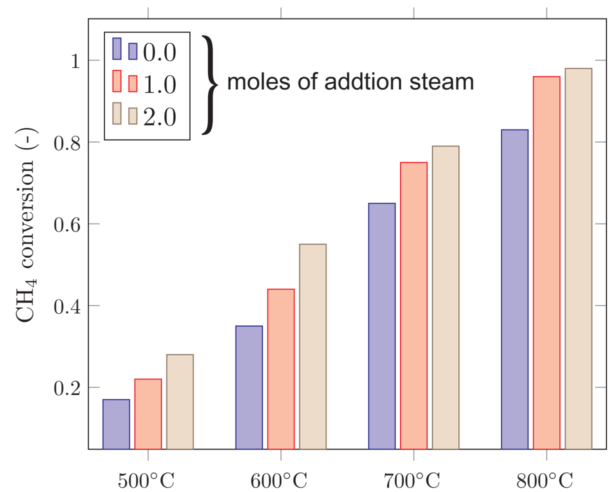

Fig. 11. Effect of the steam addition on the methane conversion at fixed CH_4: $\frac{1}{3}$ β ratio.

3.4. Effect of steam addition

The experimental setup is shown in Fig. 1, allows to investigate the effect of the steam addition on the reforming process. The study of the effect of the steam addition is an important task for the development of thermochemical waste-heat recovery systems that use the combined methane reforming with steam and flue gases.

Fig. 11 shows the changes of the methane conversion as a function of temperature when different steam content is fed into the methane reformer at a fixed CH_4: $\frac{1}{3}\beta$ ratio at 1 bar when Raschig ring is used as a catalyst. The steam addition has a significant effect on the methane conversion. The steam addition to the inlet reaction gas mixture increases the methane conversion considerably; especially at the temperature range from 500 to 800 °C. This fact is well correlated with Hegarty's [22] and Ozkara-Aydinoglu's [14] articles.

However, a constant increase of the steam amount in the inlet reaction mixture, although it increases the methane conversion, leads to a decrease in the efficiency of the reforming process. First, for the production of steam, additional heat is expended. Secondly, an increase of the mass flow of the reaction mixture through the reaction space leads to the appearance of additional aerodynamic resistance. Determination of the optimum amount of steam for the experimental scheme shown in Fig. 1, is an important task for the engineering calculations.

Steam additions have a noteworthy effect on the degree of methane conversion. The addition of extra steam to the feed stream increased CH_4 conversion considerably; especially at the temperature range from 600 °C to 800 °C which is consistent with the works of other authors [14]. They reported that the addition of steam increased methane conversion. With a further increase in the temperature range above 800 °C, it is possible to expect that the methane conversion will tend to unity for all gas compositions. This fact also has a worthy correlation with the articles of other authors [14].

The positive effect of steam addition on methane conversion can also be explained in terms of thermodynamics. It is known that the chemical system is thermodynamically favorable when its total Gibbs free energy is at its minimum value and its differential is zero for a given temperature and pressure:

$$dG^t = 0 \tag{10}$$

where G^t is total Gibbs free energy.

According to the IVTANTHERMO database [23], in the temperature range from 500 to 800 °C, a change in the Gibbs free energy provides a more likely pattern of the steam reforming reaction. For example, for temperature 500 °C ΔG of steam methane reforming is 23.4 kJ/mol and ΔG^t of dry methane reforming is 31.5 kJ/mol. Therefore, steam addition to reaction mixture significantly increases the CH_4 conversion, especially at the temperature range below 600 °C.

4. Conclusion

An experimental study of methane reforming with flue gases has been performed on Ni-Al_2O_3 catalyst over a wide range of operational parameters such as temperature, methane-to-flue gas ratio, pressure and residence time. Also, two types of catalyst shapes (solid cylinder and Raschig ring) were used for the experimental investigation.

The type of the dependence of the methane conversion as a function of temperature, residence time at various inlet gas compositions, types of catalysts, and pressures was established experimentally. For the high residence time (about 4 kg_{cat} s/mol), the methane conversion reaches a value that is close to equilibrium. The methane conversion is almost directly proportional to residence time at the temperature range from 600 to 800 °C. The catalyst shape also has an effect on the methane conversion at a constant residence time. It is established that Raschig rings give a higher methane conversion than a solid cylinder. For a wide range of operational parameters, a polynomial dependence of the methane conversion on the residence time was obtained. The statistical analysis by the Pearson criterion showed a good correlation between the experimental results and the approximating functions. The effect of pressure on the methane conversion in the pressure range from 1 to 5 bar was investigated. It is established that with increasing pressure, the methane conversion decreases. However, such a reduction is no more than 1–2% for pressures of 1 and 5 bar.

A study of the steam addition to the methane conversion was carried out. When adding steam, the methane conversion increases with fixed operating parameters. The obtained results can be used for engineering calculations of the thermochemical waste-heat recuperation systems by methane reforming with products of its complete combustion.

5. Conflict of interest

The author declare that there is no conflict of interest.

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